

Deep Learning for Search (DLS)
Summer 2026
Innopolis University
Author: Albert Nasybullin

List of Topics

Week / Pair	Topics / Activities
Week 1 / Pair 1	Introduction to Search & IR; ML System Design framework
Week 1 / Pair 2	Introduction to NLP; Diving into Tokenization; Measures of Similarity
Week 1 / Pair 3	Classical IR: TF-IDF, BM25, inverted index; Full-text Search Ranking & Rank Fusion
Week 1 / Pair 4	Ranking Metrics (MAP, MRR, nDCG, Recall@k); Lab 1; Assignment 1 released
Week 2 / Pair 1	Introduction to Deep Learning for Search
Week 2 / Pair 2	Word embeddings (word2vec, GloVe, fastText); Diving into Embeddings; Dimensionality Reduction (PCA, UMAP, t-SNE)
Week 2 / Pair 3	Dense & contextual embeddings; Contrastive learning, hard negative mining
Week 2 / Pair 4	Transformers & Attention; Assignment 1 due
Week 3 / Pair 1	Bi-encoders (DPR, Sentence-BERT)
Week 3 / Pair 2	Cross-encoders & reranking; multi-stage retrieval pipelines
Week 3 / Pair 3	Late interaction (ColBERT); Learned sparse retrieval (SPLADE); Hybrid retrieval & RRF
Week 3 / Pair 4	Learning to Rank: pointwise / pairwise / listwise, LambdaMART; Lab 2; Assignment 2 released
Week 4 / Pairs 1–2	Midterm Exam (full day)
Week 4 / Pair 3	Approximate Nearest Neighbors: HNSW, IVF, PQ; FAISS; Vector databases
Week 4 / Pair 4	Production: quantization, distillation, latency, caching; Assignment 2 due
Week 5 / Pair 1	RAG fundamentals; Chunking & document processing
Week 5 / Pair 2	Query understanding & rewriting: HyDE, query decomposition; Assignment 3 released
Week 5 / Pair 3	RAG evaluation: faithfulness, context precision/recall, RAGAS, LLM-as-judge
Week 5 / Pair 4	Agentic RAG: ReAct, Self-RAG, Corrective RAG; Lab 3
Week 6 / Pair 1	Advanced RAG: multi-hop retrieval, GraphRAG, long-context tradeoffs
Week 6 / Pair 2	Multimodal retrieval (CLIP, ColPali); end-to-end system design review; Ethics, bias, safety; Assignment 3 due
Week 6 / Pairs 3–4	Final Exam (full day)
Week 7	Project Defense

Assignments (Topics & Deadlines):

Assignment	Released	Due	Topic
Assignment 1	Week 1 / Pair 4	Week 2 / Pair 4, 23:59	Classical IR & Ranking Metrics: implementing BM25 on a corpus; computing nDCG, MAP, MRR; analysis of tokenization choices on ranking quality
Assignment 2	Week 3 / Pair 4	Week 4 / Pair 4, 23:59	Two-Stage Retrieval & Learning to Rank: building a bi-encoder → cross-encoder pipeline; hard negative mining; comparing pairwise vs listwise losses on training data
Assignment 3	Week 5 / Pair 2	Week 6 / Pair 2, 23:59	RAG System Design: assembling a RAG pipeline with multiple chunking strategies; evaluation via RAGAS-style metrics; extension to an agentic loop

Graded Activities:

Activity	Up to date	Weight in the course, %
Assignment 1	Week 2 / Pair 4, 23:59 (in class submission)	5
Assignment 2	Week 4 / Pair 4, 23:59 (in class submission)	5
Assignment 3	Week 6 / Pair 2, 23:59 (in class submission)	5
Lab 1	Week 1 / Pair 4 + 3 days, 23:59	5
Lab 2	Week 3 / Pair 4 + 3 days, 23:59	5
Lab 3	Week 5 / Pair 4 + 3 days, 23:59	5
Midterm Exam	Week 4 / Day 1	20
Final Exam	Week 6 / Day 2	30
Project Defense	Week 7	20

Grading Policy:

- Cheating zero-tolerance,
- No late submission;

Grade	%
A	(90%–100%]
B	(70%–90%]
C	[50%–70%]
D	[0%–50%)

Materials & Sources:

- “Deep Learning for Search”, Tommaso Teofili (2019)
- “Relevant Search: With applications for Solr and Elasticsearch”, Doug Turnbull and John Berryman (2016)
- “Machine Learning System Design: With end-to-end examples”, Valerii Babushkin and Arseny Kravchenko (2024)
- “Pretrained Transformers for Text Ranking: BERT and Beyond”, Jimmy Lin, Rodrigo Nogueira, Andrew Yates (2021)
- Selected papers from the course reading list (DPR, SBERT, ColBERT, SPLADE, LambdaMART, HNSW, RAG, Self-RAG, CRAG, RAGAS)